Descriptive Statistics

Key terms **KSBs: K13, S11, S52 B1**

Descriptive statistics, measures of location, central tendency, mean, median, mode, measures of spread, variance, standard deviation, discrete data, continuous data, categorical data

2.1 Types of Data

Data comes in many different types, and we can categorise it by the number of variables involved.

Definition 2.1 - Data types by number of variables

Univariate data is data with only If we have and want to study the relationship between them, we say the data is **bivariate**. We use the term **multivariate** for data with

Some data is , like categories we can classify things into. For example,

is a qualitative data type, since the outcomes could be

etc. Other data is which means it takes

Definition 2.2 - Discrete and continuous data

Discrete data is numerical data which can take
like the number of

Continuous data can take values like a measurement of

We will mostly be considering numerical data with a single variable, although the tools we are discussing today can also be applied to multivariate data.

Descriptive statistics allow us to describe sets of numerical data and their properties, without including the full data set. They can be used to summarise what data is telling us, and compare different sets of data.

2.2 Measures of Location

These are ways to describe **where** numerical data is, within the range of possible values it can take. One type that is used commonly is a **measure of central tendency**, which gives an indication of where the

Pie chart icon by diyah farida from the Noun Project

2.2.1 Mean

The mean is the most commonly used type of **average**, and is

Definition 2.3 - Mean

The **mean** of a set of values is an average found by taking the

and dividing by

For a set of n values, we write for the sum of the values, and the mean is The mean for a sample is denoted

Example 2.1.

Consider the set of values:

$$4 \quad 9 \quad 7 \quad 5 \quad 1 \quad 8 \quad 3 \quad 4 \quad 9 \quad 5 \quad 4$$

The sum of all the values is There are numbers, so Then the mean is

If there is only a finite range of values the data can take, it is easier to state the number of data points that take each value than to list them all. This kind of data is usually presented in a frequency table.

Example 2.2. *Consider the same set of values from Example 2.1. We can present this same data in a frequency table.*

.5The sum of the numbers in the frequency column is

To find the mean of the data in the table, we can and add them together to get the value of $\sum x$.

This can be divided by to get the mean of the dataset.

Value	Frequency
1	1
2	0
3	1
4	3
5	2
6	0
7	1
8	1
9	2

2.2.2 Grouped Data

When data points can take any value within a range, and there are too many values to list, we can used **grouped data**. This involves creating categories to split the data up, and then counting how many data points fall into each range.

When data is formatted in this way, we no longer have the true values of all the data points, but only which group they lie in. This means we can no longer get an exact value for the mean of this data - only an estimate.

Example 2.3.
Here is some data about the heights of children in a school year.

Height (cm)	Midpoint	Frequency (f)
60-80		10
80-100		12
100-110		21
110-120		13
120-160		4

In order to estimate the mean, we need to find the midpoint of each range of values, and use this to represent that range. We can then treat it like a normal frequency table and calculate the mean.

We multiply each and add them together to get the value of $\sum x$.

The total number of data points is the sum of the frequency column:

Then an estimate for the mean can be found by

The estimated mean height for a child in this group is

Exercise 2.1.

(a) Find the mean of these numbers:

$77 \quad 431 \quad 875 \quad 716 \quad 843 \quad 601 \quad 482 \quad 325$

(b) Find the mean of this frequency data:

x	*0*	*1*	*2*	*3*	*4*	*5*
Frequency	*15*	*34*	*94*	*45*	*17*	*35*

(c) Estimate the mean of this grouped data:

Class	*Frequency*
$0 \leq x < 10$	*42*
$10 \leq x < 20$	*47*
$20 \leq x < 30$	*94*
$30 \leq x < 50$	*35*
$50 \leq x \leq 100$	*64*

Function: AVERAGE() (E/S)

The `AVERAGE()` function (in Excel and Sheets) can be used to calculate the mean average of a data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

2.2.3 Other Measures of Location

Definition 2.4 - Median

The **median** of a dataset is an average found by taking when the values are sorted into increasing numerical order. If there are an even number of values in the set, the median is given by the (if they are not the same). The median of a sample is sometimes denoted

Example 2.4. *Consider the same set of values from Example 2.1:*

$$4 \quad 9 \quad 7 \quad 5 \quad 1 \quad 8 \quad 3 \quad 4 \quad 9 \quad 5 \quad 4$$

Sorted into order, this set is:

Of the 11 values, the so this is the median of the data.

Function: MEDIAN() (E/S)

The `MEDIAN()` function (in Excel and Sheets) can be used to calculate the median average of a data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range. If there are an even number of values in the set, it will calculate the mean average of the two in the middle.

Definition 2.5 - Mode

The **mode** of a dataset is an average found by taking

There is no standard symbol for the mode.

Example 2.5. *For the dataset above, the value appearing most frequently is a so this is the modal value. (If the data is presented in a frequency table, finding the mode just involves*

Not all datasets will have a unique mode; a data set with two modes is called and a data set with multiple modes more generally is called It is also possible for a data set to have **no mode**, in the case where all values occur

Function: MODE()/MODE.SNGL() and MODE.MULT() (E/S)

The `MODE()` or `MODE.SNGL()` function (in Excel and Sheets) can be used to calculate the mode of a data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

The `MODE.MULT()` will return multiple values if the dataset is multimodal. This will output values in the cell it is entered in, and the cells below that if needed.

Definition 2.6 - Upper and Lower Quartiles

The **upper and lower quartiles** of a data set, denoted are the values occurring

when the values are sorted into increasing numerical order. They are not averages, but describe where certain areas of the data are.

Example 2.6. *To find the upper and lower quartiles, we can*
Given our sorted data set from the Example 2.1:

$$1 \quad 3 \quad 4 \quad 4 \quad 4 \quad \mathbf{5} \quad 5 \quad 7 \quad 8 \quad 9 \quad 9$$

The median was
We can split this into two sets (excluding the median itself), and find their medians:

Then we have

(If there are an even number of values in a set, we would split it in half and find the median of each half.)

Example 2.7.
The table shows the number of kittens each of the children in a school year would like to own.

The column on the right shows the total number of kittens that the children would like to own. The mean (average) number of kittens each child wants can be calculated by dividing the total of this column

by the number of children surveyed giving

Kittens (x)	Frequency (f)	Total (xf)
0	10	
1	12	
2	21	
3	13	
4	4	
Total	60	109

To find the median of this set, we could list all 60 values and find the middle value(s):

$$0\,0\,0\,0\,0\,0\,0\,0\,0\,0\,1\,1\,1\,1\,1\,1\,1\,1\,1\,1\,1\,1\,2\,2\,2\,2\,2\,2\,2\,2\,\mathbf{2}\,\mathbf{2}\,2\,2\,2\,2\,2\,2\,2\,2\,2\,2\,2\,3\,3\,3\,3\,3\,3\,3\,3\,3\,3\,3\,3\,3\,4\,4\,4\,4$$

In this case, the median value is To find this from the table, we must first note that the total number of values is so we are looking for the

Similarly, we can determine that the quartiles will be the

so they will be

Function: QUARTILES (E/S)

The `QUARTILE` function (in Excel and Sheets) can be used to calculate the quartiles of a data set. It takes two inputs: a range of values, and a number to indicate which quartile you want (0, 1, 2, 3 or 4). Choosing 0 will output the minimum value, 2 will output the median, and 4 will output the maximum value. It will ignore any non-numerical data included in the range.

Exercise 2.2.

(a) For each of the datasets in Exercise 2.1, find the mode (or modal category), and find or estimate the median and upper/lower quartiles.

(b) For the children's heights data from Example 2.3, find the modal category and an estimate of the upper and lower quartiles.

2.3 Measures of Spread

As well as measures of location, we can use statistics to describe the **spread** of a data set - describing how spread out the data are.

Definition 2.7 - Range

The **range** of a data set is given by the

Example 2.8. *In the example data set, the largest value is* *so the range is*

Definition 2.8 - Interquartile Range

The **interquartile range** of a dataset is the

Example 2.9. *In our earlier example, we had* *Then the interquartile range, or*

Function: Range and IQR (E/S)

The `MAX()` and `MIN()` functions (in Excel and Sheets) can be used to find the maximum and minimum values in a dataset, and the difference between them (`MAX() - MIN ()`) will tell you the range. You will need to give it a range of cells, which will all need to contain numerical values. (The `MAXA()` and `MINA()` functions will do the same thing, ignoring any non-numerical values in the range).

The IQR of a dataset can similarly be calculated using the `QUARTILE()` function, by calculating (e.g.) `QUARTILE(A1:A10, 3) - QUARTILE(A1:A10, 1)`.

Exercise 2.3.

(a) For each of the datasets in Exercise 2.1, find or estimate the range and interquartile range.

(b) For the children's heights data from Example 2.3, using the quartiles you calculated earlier, find the interquartile range.

These measures tell us a little about how spread out the data are, but they don't use the whole data set, so can only capture some of the information. We have more powerful tools, which tell us more about the data.

2.3.1 Variance

Definition 2.9 - Variance

The **variance** of a sample, denoted is a measure of how spread out the data are -

- and for a set of n values it can be calculated as:

This formula says that for each data point in the sample, we find

then take the square of that value. Then we add these values together and divide by (so, if there are n values, we divide by the number one less than that).

Example 2.10. *Consider the sample:*

$$1 \quad 6 \quad 9 \quad 4 \quad 8 \quad 6$$

The mean of this set is

Subtract this number from each data point:

$$-4.66\ldots \qquad 0.33\ldots$$

Then square each and take the sum:

$$(-4.66\ldots)^2 + (0.33\ldots)^2 + (3.33\ldots)^2 + (-1.66\ldots)^2 + (2.33\ldots)^2 + (0.33\ldots)^2 =$$

Finally, divide by $n-1$:

So the variance is

Note

You might wonder why we square the differences from the mean in calculating the variance. If we were to just take the differences, some would be positive and others negative - and if we added them all together Squaring all the differences makes them and also improves the usefulness of variance as a measure.

We can also calculate the variance using a similar (but not the same!) formula:

Instead of calculating the differences between each value and the mean, we are now taking the difference between the

Example 2.11.

Consider the same sample as in Example 2.10.

$$1 \qquad 6 \qquad 9 \qquad 4 \qquad 8 \qquad 6$$

The sum of the values is *The sum of the squares is given by:*

$$\sum x^2 = 1^2 + 6^2 + 9^2 + 4^2 + 8^2 + 6^2 =$$

So using the formula:

This matches the previous value for the variance.

It's possible to show that these two formulae are equivalent. The second form is in general easier to calculate.

Exercise 2.4.

For each of the datasets in Exercise 2.1, find or estimate the variance.

Function: VAR.S (E) / VAR (S)

The `VAR.S()` function (in Excel) or the `VAR()` function (in Google Sheets) can be used to calculate the variance of a sample data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

The variance does give a measure of how spread-out the data is, but it's less useful for meaningful comparisons. A more useful measure, and a fundamental idea in statistics, is the **standard deviation**.

2.3.2 Standard deviation

Definition 2.10 - Standard Deviation

The **standard deviation** of a sample, denoted is the

of that sample. It can be calculated as:

Roughly, the standard deviation tells us the

Function: STDEV.S (E) or STDEV (S)

The `STDEV.S()` function (in Excel) and the `STDEV()` function (in Google Sheets) can be used to calculate the standard deviation of a sample. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

Example 2.12.

In Example 2.10, we calculated the variance of a sample to be 8.27. *To find the standard deviation, we can take the square root to get*

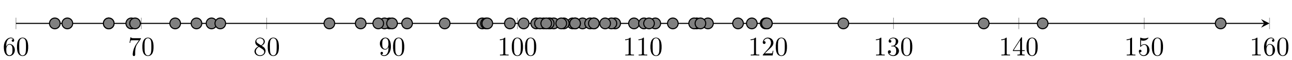
Exercise 2.5.

(a) For each of the datasets in Exercise 2.1, find or estimate the standard deviation.

(b) For the children's heights data from Example 2.3, find an estimate of the standard deviation.

2.3.3 Outliers

This plot shows the heights of the children from Example 2.3.



The mean of the data was estimated to be and we estimated the quartiles to be
The IQR was estimated to be The standard deviation of this data was

While most of the data is clustered together in an area just higher than the mean, there are some data points which lie further away. Data points like this are called **outliers**, and many types of data will include them.

Definition 2.11 - Outliers

We can say a data point is an **outlier** if it lies more than or alternatively away from the mean.

This is a widely-used definition, although it is not universal. Outliers can be due to false readings, and are sometimes excluded from datasets to facilitate analysis - although this should be done with care, especially if it might affect the validity of the results.

Exercise 2.6.

Which of the data points on the plot above are outliers, by the IQR definition, and by the standard deviation definition?

2.3.4 Limitations of Summary Statistics

For each of the measures we have seen, there are some advantages and disadvantages.

Mean, Standard Deviation and Variance These three numerical measures are commonly used for symmetrical data (data which is equally spread out around the middle), or for large sets of data (e.g. n=1000). Typically, the standard deviation and variance are used in conjunction with the mean.

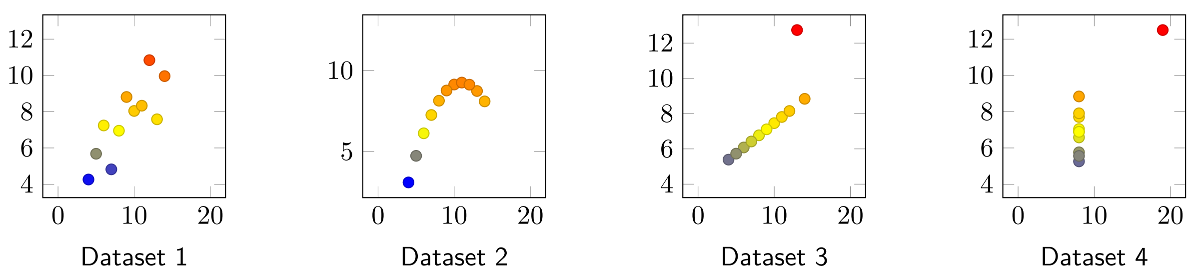
These measures use all the data values, giving a true indication of that measure, and each data value equally influences the measure being calculated. However, in a small sample, any extreme values may influence and skew the measure.

Median and Inter-quartile Range The median and interquartile range are commonly used with smaller sets of data, or data which are skewed to one side (e.g. more at the higher end than the lower end). Typically, the median is used in conjunction with the inter-quartile range.

They are not affected by extreme values, and can be calculated from incomplete data sets. However, these measures don't take into account all the values in the data. Without using technology, if the data set is large, the median may be difficult to locate.

Mode and Range The mode and range are commonly used when the data is incomplete or unknown. While they are both relatively easy to calculate, data sets may be multimodal, or have no mode - and depending on the context, the mode may not have any practical meaning. The range is also heavily influenced by extreme values.

2.3.5 Anscombe's Quartet



These plots show a collection of datasets known as **Anscombe's Quartet**. The summary statistics for these datasets are given in the table.

Even though the data sets look different, they have

in both the x and y directions. The data have been carefully chosen to have this property; it doesn't mean we shouldn't trust these measures, but we need to be careful not to rely on them too much.

Dataset	**1**	**2**	**3**	**4**
Mean x				
Mean y (2 d.p)				
s^2 x				
s^2 y ($\pm$0.003)				

2.4 Populations and Samples

Definition 2.12 - Population and Sample

The **population** is the
Populations are generally too large to measure every subject, but are usually what we're interested in learning about. A **sample** is a smaller collection that is

which we can study in order to make inferences about the population.

Later we will see how to choose samples, and how to use the statistics we can calculate from the samples to infer things about the wider population.

The statistics we have seen use slightly different notation for the sample and the whole population.

Definition 2.13 - Notation for samples and populations

We use $\bar{x}$ to denote the sample mean, s_x^2 for the sample variance and s_x for the sample standard deviation.

When studying populations, it is more conventional to use the Greek letter to denote

the population mean, and to write for the population variance and for the population standard deviation. This is the Greek letter which is the lower-case version of

The calculation for the variance and standard deviation of a population is slightly different - instead of dividing by $n-1$, so the formulae become:

Function: VAR.P and STDEV.P (E) / VARP and STDEVP (S)

The `VAR.P()` function (in Excel) or the `VARP()` function (in Google Sheets) can be used to calculate the variance of a population data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

The `STDEV.P()` function (in Excel) or the `STDEVP()` function (in Google Sheets) can be used to calculate the standard deviation of a population data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.